

Introduction of a new variable

- Consider the problem (**P**),

$$\text{Min } \mathbf{c}^T \mathbf{x}$$

subject to

$$\mathbf{Ax} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}.$$

Let $\mathbf{x}_0$ be an **optimal BFS** of this problem.

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- A column is added to the matrix A call it $\tilde{\mathbf{a}}_{n+1}$ and a component c_{n+1} is added to the cost vector $\mathbf{c}$, which is the cost associated with the new variable x_{n+1} .

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- Min $[\mathbf{c}, c_{n+1}]^T \mathbf{x}_{(n+1) \times 1}$
subject to
 $[\mathbf{A} : \tilde{\mathbf{a}}_{n+1}] \mathbf{x}_{(n+1) \times 1} = \mathbf{b}, \quad \mathbf{x}_{(n+1) \times 1} \geq \mathbf{0}.$

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where $\mathbf{a}_{B,m+1}^T$ and $\mathbf{a}_{N,m+1}^T$ are the components of $\mathbf{a}_{m+1}^T$ corresponding to the basic variables and nonbasic variables of $\mathbf{x}_0$, respectively.

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- The **inverse** of this new basis matrix is given by

$$\begin{bmatrix} B^{-1} & \mathbf{0} \\ -\mathbf{a}_{B,m+1}^T B^{-1} & 1 \end{bmatrix}.$$

- The new row added to the (main) simplex table will be of the form

$$\begin{aligned} & [-\mathbf{a}_{B,m+1}^T B^{-1}, 1] \begin{bmatrix} \mathbf{a}_{B,m+1}^T B & \mathbf{a}_{N,m+1}^T N & \mathbf{0} \\ \mathbf{a}_{B,m+1}^T B^{-1} & \mathbf{a}_{N,m+1}^T N^{-1} & 1 \end{bmatrix} = \\ & [\mathbf{0}, -\mathbf{a}_{B,m+1}^T B^{-1} N + \mathbf{a}_{N,m+1}^T, 1] \end{aligned}$$

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- The new RHS becomes

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- Since the cost associated with s_{m+1} is equal to zero, all the $c_j - z_j$ entries remain unchanged and are non negative.
- Then **Dual Simplex** can be used either to obtain the new optimal solution or to conclude that new (P) is **infeasible**.

Changing entries in the coefficient matrix A :

- **Case 1:** The column corresponding to a **nonbasic variable** say $\tilde{a}_j$ of the optimal solution is changed to $\tilde{a}'_j$.

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- If the new $c_j - z'_j$ value satisfies the **optimality condition**, then the previous optimal solution will be **optimal** for the new problem.

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- If **not**, then use **Simplex** either to obtain the new **optimal solution** or to conclude that new (P) does **not** have an optimal solution.

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- If $u_{j,(n+1)} = 0$, then it implies that $\{\tilde{\mathbf{a}}_1, \dots, \tilde{\mathbf{a}}_{j-1}, \tilde{\mathbf{a}}'_j, \tilde{\mathbf{a}}_{j+1}, \dots, \tilde{\mathbf{a}}_m\}$ is **LD** and we have to find an **initial BFS** for the changed problem.

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- If $u_{j,(n+1)} \neq 0$, then pivot on this element and make x_{n+1} enter the basis and x_j leave the basis.

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- The necessary calculations mentioned above might disturb the optimality as well as the feasibility of the current **BFS**.
- If however the new $c_j - z_j$ values are all **nonnegative** and all the RHS entries remain **non negative**, then the table is **optimal** and the corresponding BFS is **optimal** for the new **(P)**.

- If the new table has all RHS entries **nonnegative**, but at least one of the $c_j - z_j$ values is **negative** then use **Simplex** either to obtain the new optimal solution, or to conclude that the new **(P)** **does not** have an optimal solution.

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- If all the $c_j - z_j$ values are **nonnegative** but at least one of the RHS entries in the simplex table is **negative** then use **Dual Simplex** to obtain the new optimal solution or to conclude that the new **(P)** has **no** feasible solution.

Artificial Variable Method to find an initial BFS of an LPP: Big- M method:

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- An initial BFS is $[\mathbf{0}_{1 \times n}, b_1, \dots, b_m]^T$ with basic variables $w_1, \dots, w_m$.
- $\mathbf{x} \in \text{Fea}(\mathbf{P}) \Leftrightarrow [\mathbf{x}^T, \mathbf{0}_{1 \times m}]^T \in \text{Fea}(\text{LP}(\mathbf{M}))$.

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- **Case 1a:** $[\mathbf{x}_*^T, \mathbf{0}_{1 \times m}]^T$ is an **optimal** solution of $\text{LP}(\mathbf{M})$ with all the artificial variables taking 0 value.
Then $\mathbf{x}_*$ is an **optimal** solution for $\mathbf{P}$.

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- **Case 2:** $\text{LP}(\mathbf{M})$ **does not** have an **optimal** solution.

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Then $\mathbf{P}$ does **not** have a **feasible** solution.
- The above conclusion of **Case 2b** that $\mathbf{P}$ is **infeasible** may not be true if $c_k - z_k$ is **not** the **most negative** among the $c_j - z_j$ values.

- **Case 2(summarized):** In some iteration (or simplex table), there exists a k such that $c_k - z_k < 0$, the corresponding column $B^{-1}\tilde{\mathbf{a}}_k \leq \mathbf{0}$ for **LP(M)**.

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- **Remark 1:** If at some stage of solving the artificial variables **leave** the basis, then the corresponding **BFS** is a **BFS** of **P**.

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- **Remark 1:** If at some stage of solving the artificial variables **leave** the basis, then the corresponding **BFS** is a **BFS** of **P**. Then **delete** all columns corresponding to the **artificial variables** and continue.

- **Example 1** : Consider the problem **P**,

Minimize $x_1 - x_2$

subject to

$$2x_1 + x_2 \geq 4$$

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We do **not** have a **feasible solution** of the **dual** of **P** at hand.
- The (**Big-M**) is used which provides an **initial BFS** of **P** for Simplex algorithm.

- Consider the modified problem

Minimize $x_1 - x_2 + Mw$

subject to

$$2x_1 + x_2 - s_1 + w = 4$$

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- The initial table corresponding to the basic variables w and s_2 is given below.

$c_j - z_j$	$1 - 2M$	$-M - 1$	M	0	0	
	$B^{-1}\tilde{a}_1$	$B^{-1}\tilde{a}_2$	$B^{-1}s_1$	$B^{-1}s_2$	$B^{-1}w$	$B^{-1}b$
w	2	1	-1	0	1	4
s_2	1	-1	0	1	0	1

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w	2	1	-1	0	1	4
s_2	1	-1	0	1	0	1

- x_1 will be the **entering variable** and s_2 will be the **leaving variable** for the next table.



$c_j - z_j$	0	$-3M$	M	$2M - 1$	0	
	$B^{-1}\tilde{\mathbf{a}}_1$	$B^{-1}\tilde{\mathbf{a}}_2$	$B^{-1}\mathbf{s}_1$	$B^{-1}\mathbf{s}_2$	$B^{-1}\mathbf{w}$	$B^{-1}\mathbf{b}$
w	0	3	-1	-2	1	2
x_1	1	-1	0	1	0	1



$c_j - z_j$	0	$-3M$	M	$2M - 1$	0	
	$B^{-1}\tilde{\mathbf{a}}_1$	$B^{-1}\tilde{\mathbf{a}}_2$	$B^{-1}\mathbf{s}_1$	$B^{-1}\mathbf{s}_2$	$B^{-1}\mathbf{w}$	$B^{-1}\mathbf{b}$
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- x_2 will be the **entering variable** and the artificial variable w will be the **leaving variable**.



$c_j - z_j$	0	$-3M$	M	$2M - 1$	0	
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x_2	0	1				$\frac{2}{3}$
x_1	1	0				$1 + \frac{2}{3}$

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We consider the corresponding modified problem

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$c_j - z_j$	$1 - 3M$	$1 - 5M$	0	M	0	
	$B^{-1}\tilde{\mathbf{a}}_1$	$B^{-1}\tilde{\mathbf{a}}_2$	$B^{-1}\mathbf{s}_1$	$B^{-1}\mathbf{s}_2$	$B^{-1}\mathbf{w}$	$B^{-1}\mathbf{b}$
s_1	1	2	1	0	0	2
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$c_j - z_j$	$\frac{1}{2}(1 - M)$	0	$\frac{1}{2}(5M - 1)$	M	0	
	$B^{-1}\tilde{\mathbf{a}}_1$	$B^{-1}\tilde{\mathbf{a}}_2$	$B^{-1}\mathbf{s}_1$	$B^{-1}\mathbf{s}_2$	$B^{-1}\mathbf{w}$	$B^{-1}\mathbf{b}$
x_2	$\frac{1}{2}$	1	$\frac{1}{2}$	0	0	1
w	$\frac{1}{2}$	0	$-\frac{5}{2}$	-1	1	10

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$c_j - z_j$	0	$M - 1$	$3M - 1$	M	0	
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x_1	1	2	1	0	0	2
w	0	-1	-3	-1	1	9

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- Since the above optimal table has an artificial variable taking positive value hence the original problem (without the artificial variable) has no feasible solution.

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The initial table is given by:

$c_j - z_j$	1	1	0	0	
	$B^{-1}\tilde{\mathbf{a}}_1$	$B^{-1}\tilde{\mathbf{a}}_2$	$B^{-1}\mathbf{e}_1$	$B^{-1}\mathbf{e}_2$	$B^{-1}\mathbf{b}$
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Here s_2 is the **leaving variable** and x_2 the **entering variable**.

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$c_j - z_j$	$\frac{2}{5}$	0	0	$\frac{1}{5}$	
	$B^{-1}\tilde{\mathbf{a}}_1$	$B^{-1}\tilde{\mathbf{a}}_2$	$B^{-1}\mathbf{e}_1$	$B^{-1}\mathbf{e}_2$	$B^{-1}\mathbf{b}$
s_1	$-\frac{1}{5}$	0	1	$\frac{2}{5}$	-4
x_2	$\frac{3}{5}$	1	0	$-\frac{1}{5}$	3

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x_1	1	0	-5	-2	20
x_2	0	1	3	1	-9

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	$B^{-1}\tilde{\mathbf{a}}_1$	$B^{-1}\tilde{\mathbf{a}}_2$	$B^{-1}\mathbf{e}_1$	$B^{-1}\mathbf{e}_2$	$B^{-1}\mathbf{b}$
s_1	$-\frac{1}{5}$	0	1	$\frac{2}{5}$	-4
x_2	$\frac{3}{5}$	1	0	$-\frac{1}{5}$	3

- Here s_1 is the leaving variable and x_1 the entering variable. The next table is given by:

$c_j - z_j$	0	0			
	$B^{-1}\tilde{\mathbf{a}}_1$	$B^{-1}\tilde{\mathbf{a}}_2$	$B^{-1}\mathbf{e}_1$	$B^{-1}\mathbf{e}_2$	$B^{-1}\mathbf{b}$
x_1	1	0	-5	-2	20
x_2	0	1	3	1	-9

Hence clearly the problem is infeasible.